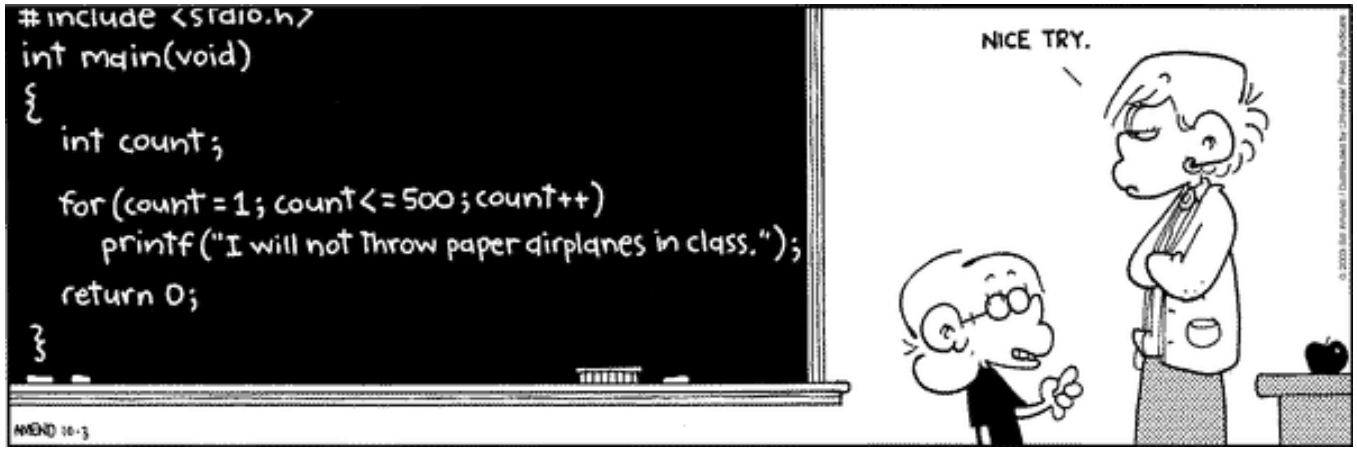
Admin

- Assign 2 grading results out soon, revise and resubmit on open issues
- printf perseverance and pride!!



Today: Thanks for the memory!

Runtime stack, stack frame layout

Linker memory map, address space layout

Loading, how an executable file becomes a running program

Heap allocation, malloc and free

```
// start.s
lui   sp,0x50000
call  main
```

```
void main(void)
{
    delta(3, 7);
}

int delta(int a, int b)
{
  int diff = sqr(a) - sqr(b);
  return diff;
}

int sqr(int v)
{
  return v * v;
}
```

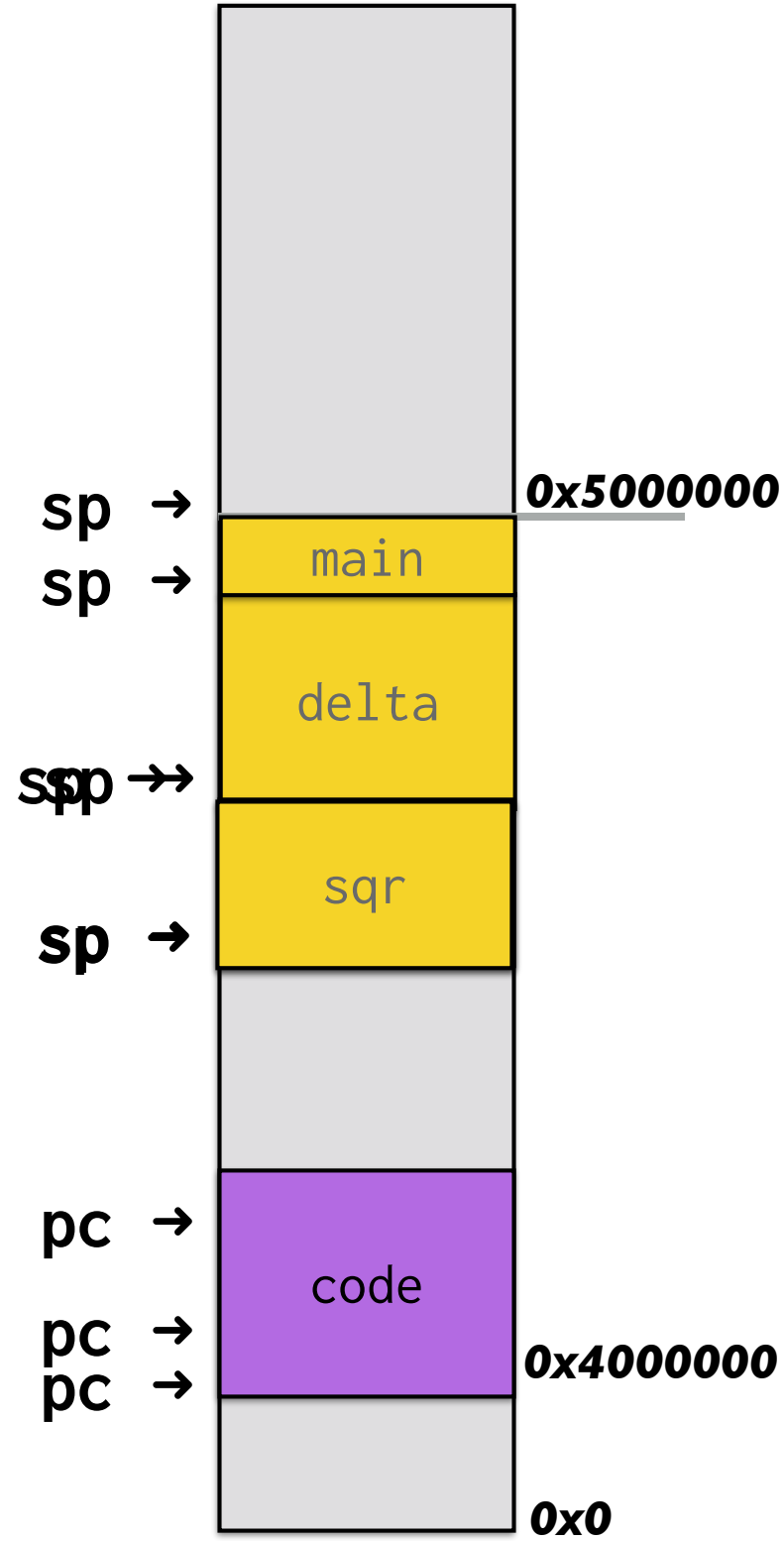
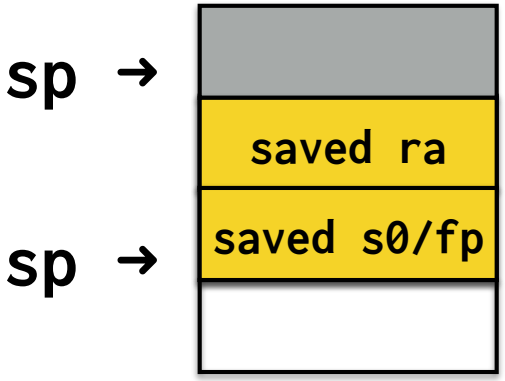


Diagram not to scale

Stack operation

```
addi   sp,sp,-16
sd     ra,8(sp)
sd     s0,0(sp)
addi   s0,sp,16
mv     a1,a0
call   sum
ld     ra,8(sp)
ld     s0,0(sp)
add    sp,sp,16
ret
```



Frame pointer

Conventions for frame pointer and stack frame layout

Enable reliable stack introspection

CFLAGS to enable: `-f-no-omit-frame-pointer`

`s0` used as `fp`

Adds prolog/epilog to each function that sets up/tears down the standard frame and manages `fp`

Trace stack frames

Prolog

Adjust stack pointer to make space (16 + `rest`)

Store ra and fp to stack

Set `fp` to where sp was (end of prev stack frame)

Body

`fp` stays anchored

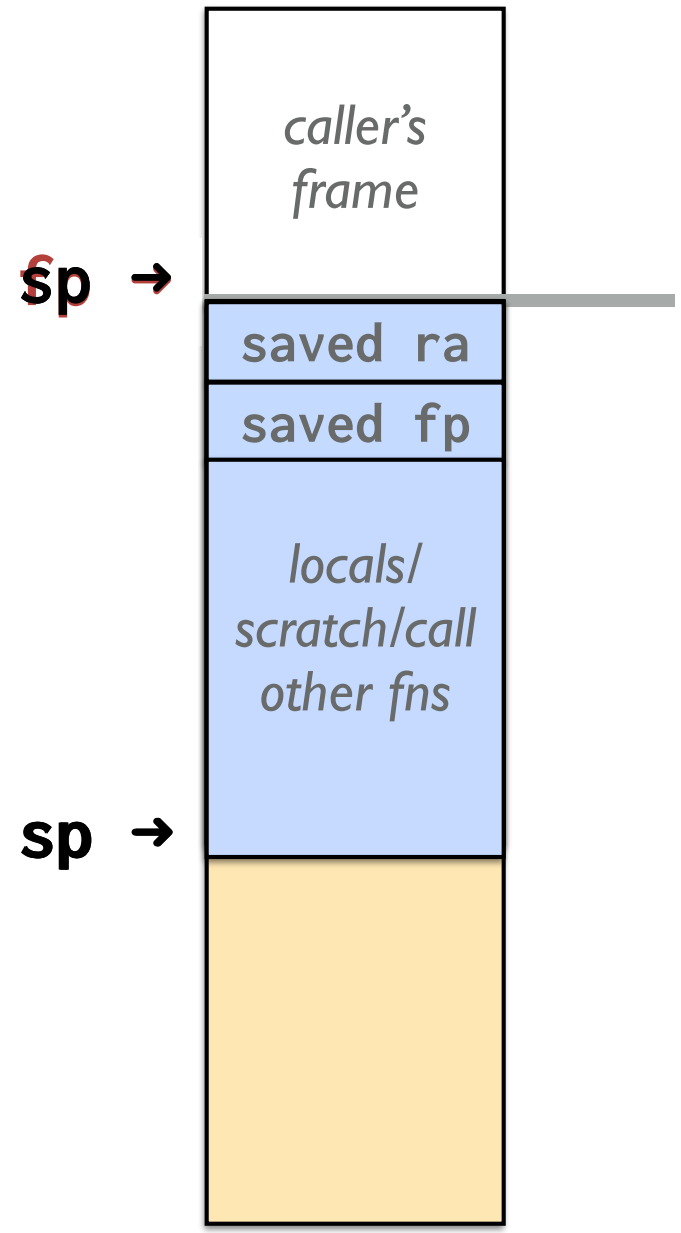
access data on stack `fp`-relative

fp offsets remain stable (even if **sp** changes)

Epilog

Restore ra, fp to saved values

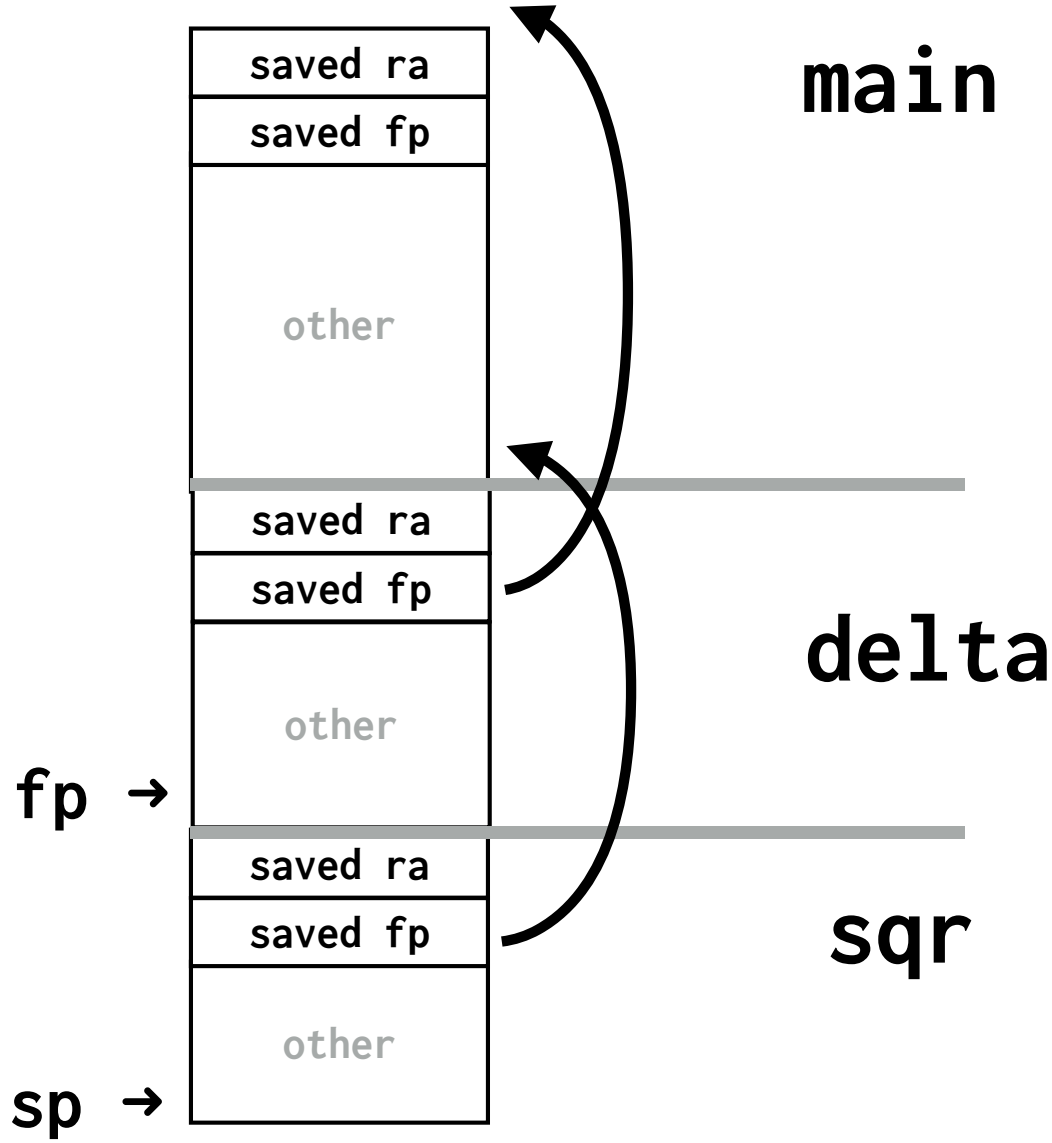
Adjust sp to remove frame



Frame pointers form linked chain

Can start at currently executing call (**sqr**) and back up to caller (**delta**), from there to its caller (**main**), who ends the chain

```
// start.s

// init fp = 0 as termination
li fp,0
lui sp,0x50000
jal main
```



other =
more saved regs, locals, scratch

Deep dive into full frame coming up in this week's lab!

Frame pointer Pros/Cons

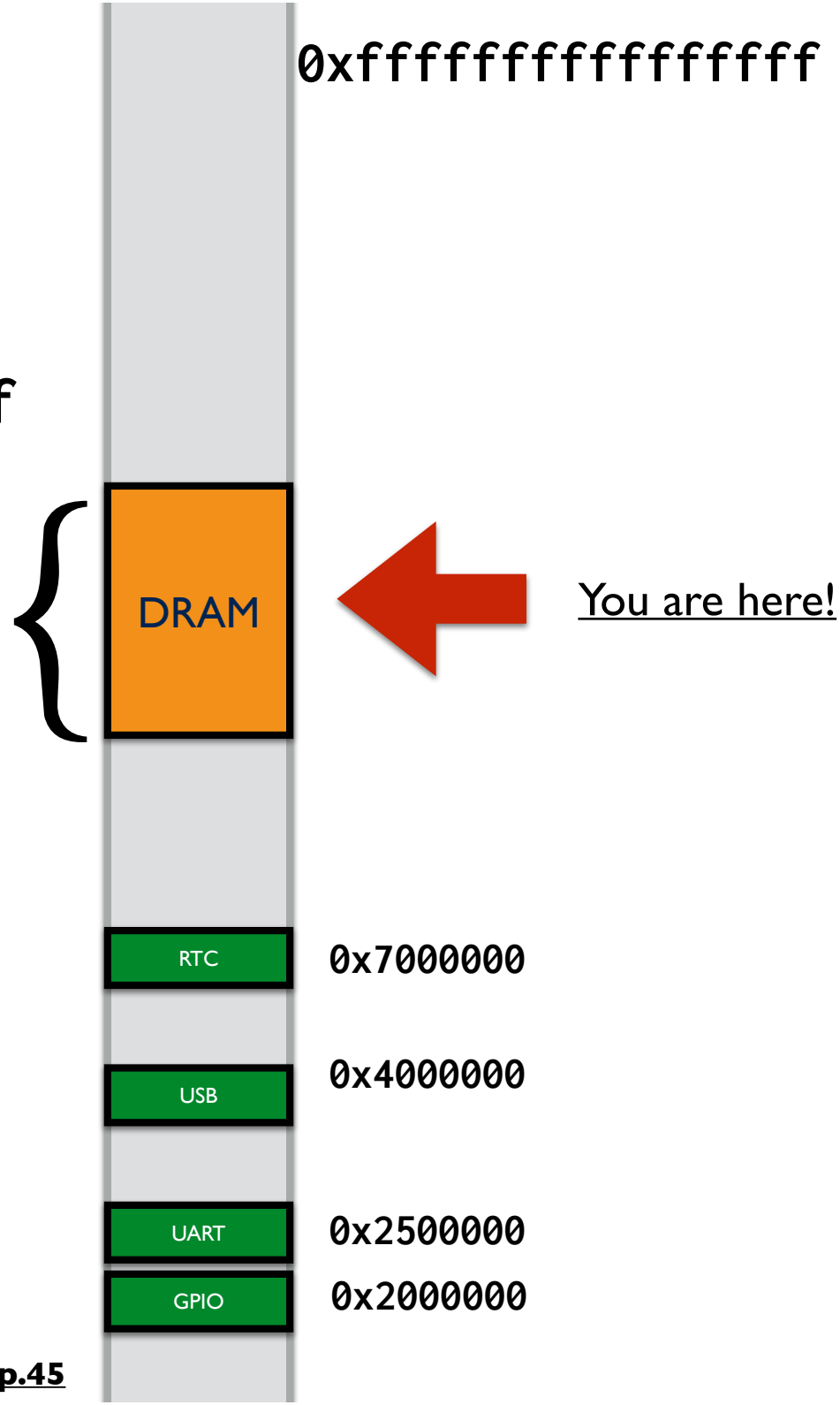
+ Anchored `fp`, offsets are constant
+ Standard frame layout enables runtime introspection
+ Backtrace for debugging
+ Unwind stack on exception

- Fixed overhead cost, every function call affected
- Adds 2-4 instructions to setup/tear down frame
- Adds 8-16 bytes to frame size

Memory Map

64-bit address space
Addresses 0 to 0xffffffff

1 GB of physical RAM
0x40000000-0x7fffffff



Ref: D1-H User Manual p.45

```
SECTIONS
{
   .text 0x40000000 :{ *(.text.start)
                       *(.text*)}
   .rodata :          { *(.rodata*) }
   .data :            { *(.data*) }
   __bss_start        = . ;
   .bss :             { *(.bss*) }
   __bss_end          = . ;
}
```

Use this memory for heap☞

(zeroed data) .bss

(initialized data) .data

(read-only data) .rodata

.text

0x50000000

```
_start:
   li    fp,0
   lui   sp,0x50000
   jal   _cstart
```

```
void _cstart(void) {
  char *bss = &__bss_start;
  while (bss < &__bss_end)
     *bss++ = 0;
  }
  main();
}
```

__bss_end

__bss_start

0x40000000

blink.bin

```
$ xfel write 0x40000000 blink.bin
$ xfel exec 0x40000000
```

We have global storage ...

+ **Convenient**

  Fixed location, shared across entire program

  No explicit allocate/deallocate

+ **Fairly efficient, plentiful**

  (But cost to send over serial line to bootloader)

+/- **Scope and lifetime is global**

  No encapsulation, hard to track use/dependencies

  One shared namespace, possibility of conflicts

  Frowned upon stylistically

... and stack storage ...

+ **Convenient**

  Automatic alloc/dealloc on function entry/exit

+ **Efficient, fairly plentiful**

  (But finite size limit on total stack usage)

+/- **Scope/lifetime dictated by control flow**

  Private to stack frame

  Does not persist after function exits

Why do we also need a heap?

An example:

`code/heap/names.c`

Dynamic storage

+ Programmer controls scope/lifetime

Versatile, precise

Works for situations where global/stack do not

- Needs software runtime support

Library routines manage the heap memory and

Process allocation/deallocation requests

- C version is low on safety

No type safety (raw void *, number of bytes)

Much opportunity for error

(allocate wrong size, use after free, double free)

Heap interface

```
void *malloc (size_t nbytes);
void  free (void *ptr);
```

void* pointer

"Generic" pointer, a memory address

Type of pointee is not specified, could be any data

What you can do with a void*

Pass to/from function, pointer assignment

What you cannot do with a void*

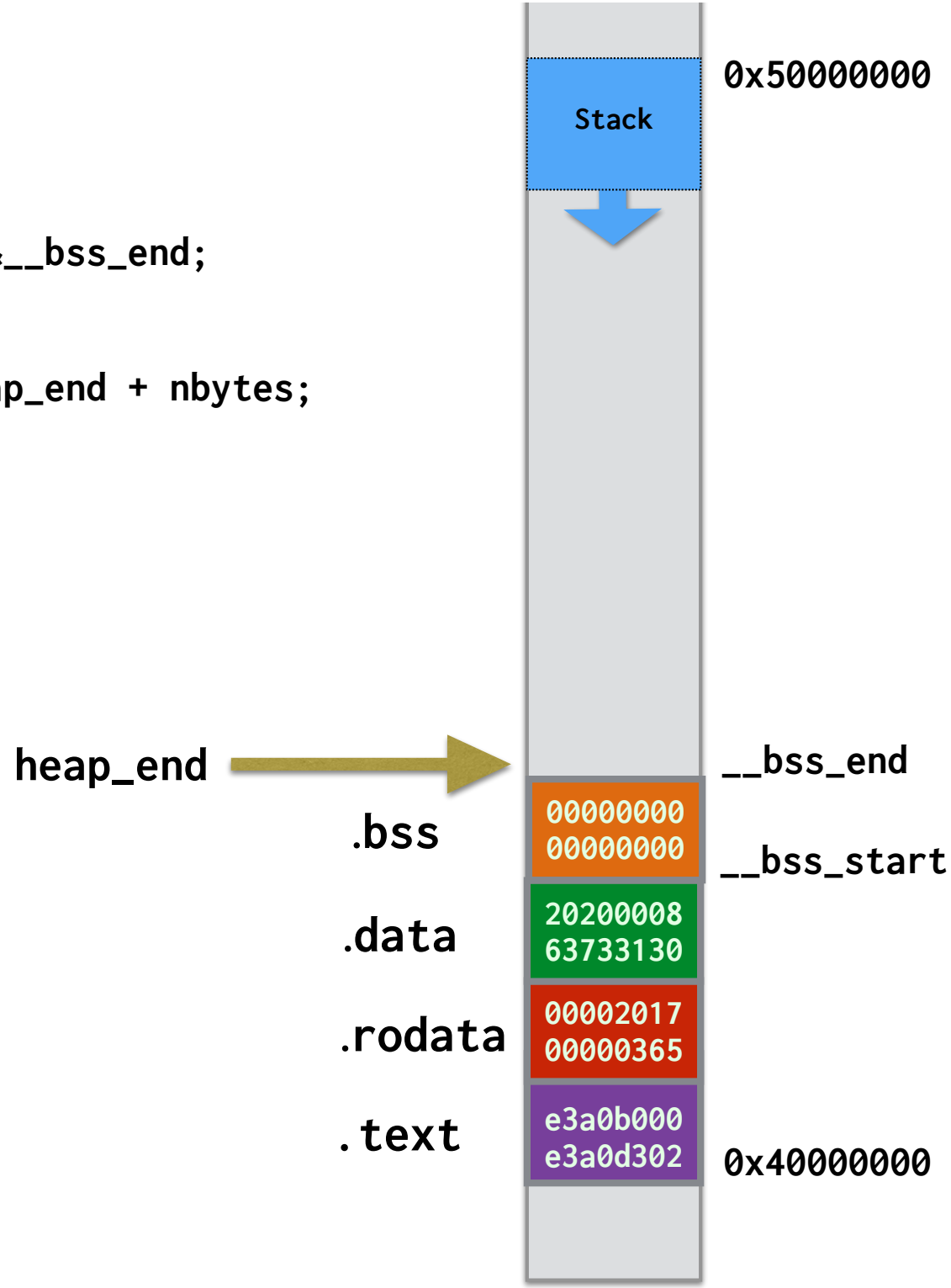
Cannot dereference (must cast first)

Cannot do pointer arithmetic (cast to char * to manually control scaling)

Cannot use array indexing (size of pointee not known!)

How to implement a heap

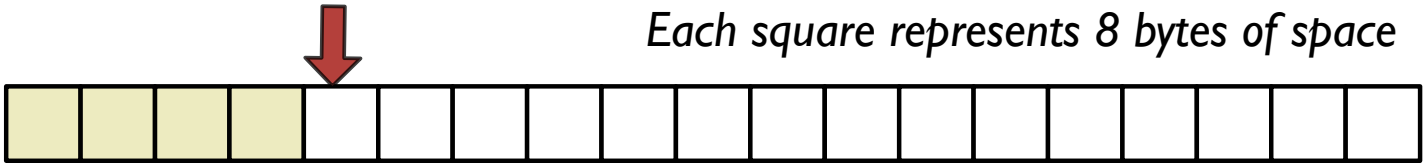
```
void *sbrk(int nbytes)
{
    static void *_cur_heap_end = &__bss_end;

    void *prev_end = cur_heap_end;
    cur_heap_end = (char *)cur_heap_end + nbytes;
    return prev_end;
}
```



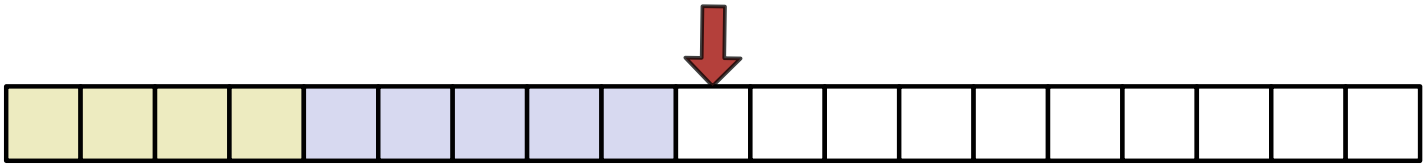
Tracing the bump allocator

Each square represents 8 bytes of space

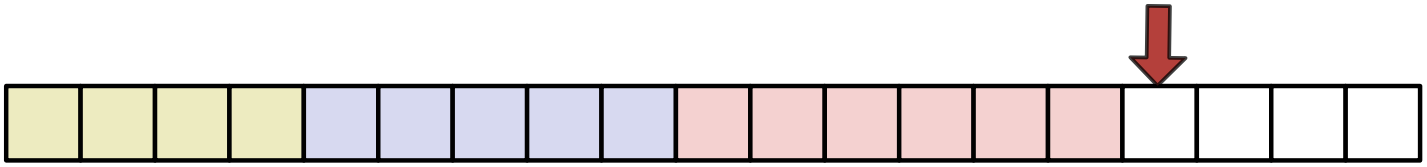
```
p1 = malloc(32)
```



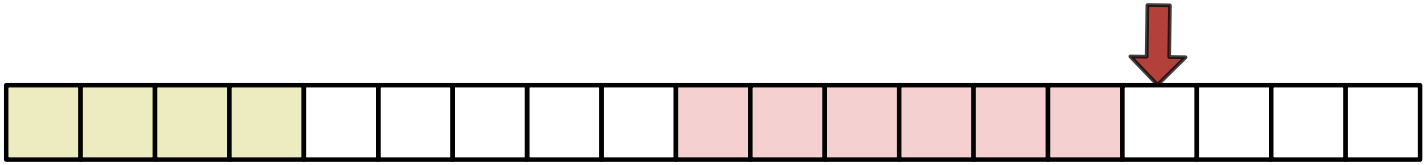
```
p2 = malloc(40)
```



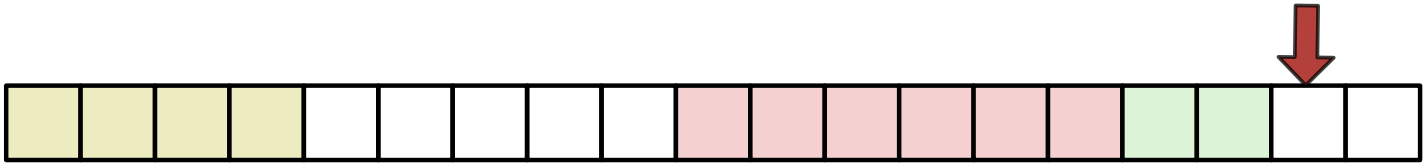
```
p3 = malloc(48)
```



```
free(p2)
```



```
p4 = malloc(16)
```



Bump Memory Allocator

`code/heap/malloc.c`

Evaluate bump allocator

+ Operations super-fast

+ Very simple code, easy to verify, test, debug

- No recycling/re-use

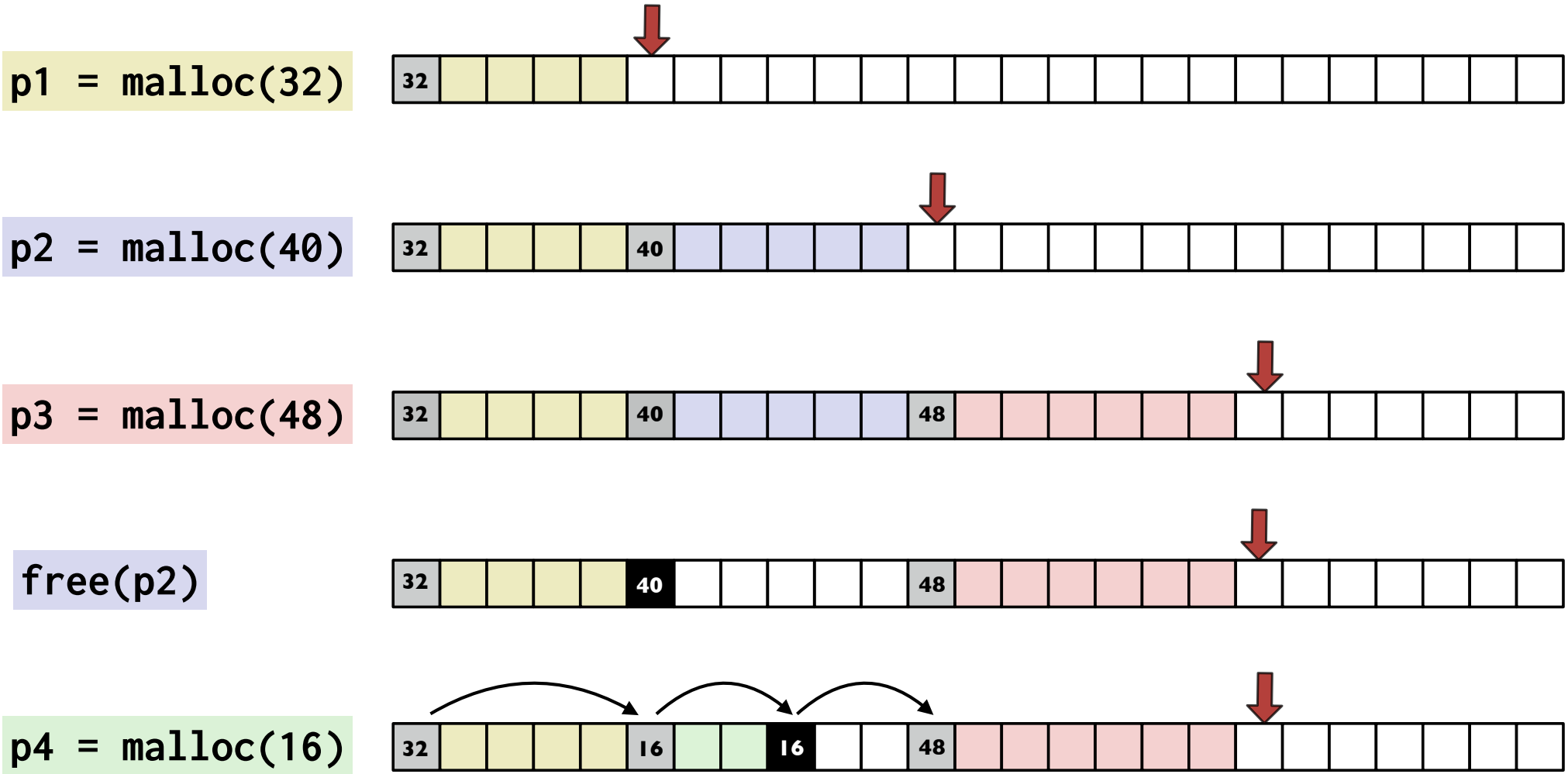
  (in what situations will this be problematic?)

- Sad consequences when `sbrk()` advances into stack

  (what can we do about that?)

Pre-block header, implicit list

Each square represents 8 bytes, header records size of payload in bytes



Header struct on each block

```
struct header {
    unsigned int size;
    unsigned int status;
};                              // sizeof(struct header) = 8 bytes

enum { IN_USE = 0, FREE = 1};


void *malloc(size_t nbytes)
{
    nbytes = roundup(nbytes, 8);
    size_t total_bytes = nbytes + sizeof(struct header);

    struct header *hdr = sbrk(total_bytes); // extend end of heap
    hdr->size = nbytes;
    hdr->status = IN_USE;
    return hdr + 1;     // return address at start of payload
}
```

Challenges for malloc client

- **Correct allocation (size in bytes)**
- **Correct access to block (within bounds, not freed)**
- **Correct free (once and only once, at correct time)**

What happens if you…

— forget to free a block after you are done using it?
— access a memory block after you freed it?
— free a block twice?
— free a pointer you didn't malloc?
— access outside the bounds of a heap-allocated block?

Challenges for malloc implementor

just `malloc` is easy 😎
`malloc` with `free` is hard 🤔
Efficient `malloc` with `free` ….Yikes! 🥵

Complex code (pointer math, typecasts)
Thorough testing is challenge (more so than usual)
Critical system component
- correctness is non-negotiable!

Survival strategies:
- draw pictures
- printf (you've earned it!!)
- early tests use examples small enough to trace by hand if need be
- build up to bigger, more complex tests